Physics Horizons — Solution

X25 (2025) — English translation

A1 — 0.20 pts

Let the side of the square hole be $a = 1$ m. Find the characteristic angular dimension β of the hole as viewed from the screen. Compare it to the angular size of the Sun.

Let's calculate the angular size of the side of the square. Since $a \ll L$, then:

$$\beta \approx \frac{a}{L}.$$

Answer:

$$\beta \approx 6^\circ$$

A2 — 0.20 pts

Now let the side of the square hole be $a = 5$ mm. Find the characteristic angular dimension β of the hole as viewed from the screen. Compare it to the angular size of the Sun.

Similar to the previous point:

$$\beta \approx \frac{a}{L}.$$

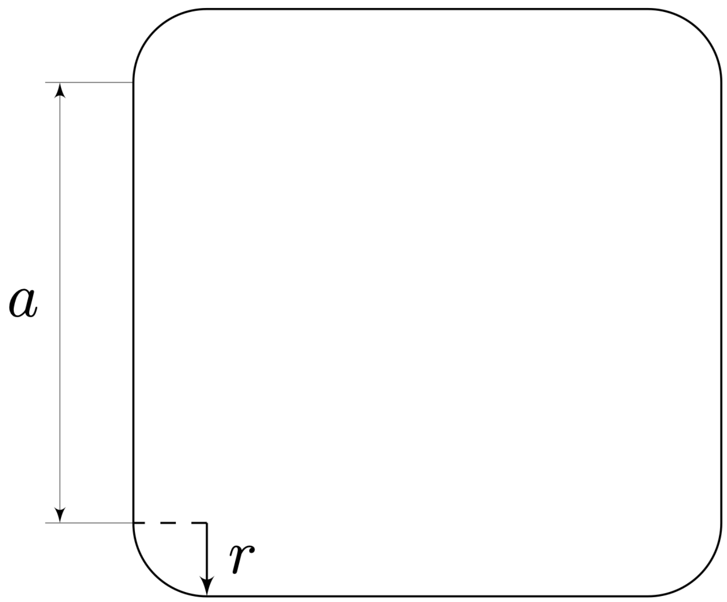
Answer:

$$\beta \approx 0.03^\circ$$

A3 — 0.30 pts

In the answer sheets, qualitatively give the image of the Sun at $a = 1$ m. Please indicate the dimensions that characterize this image.

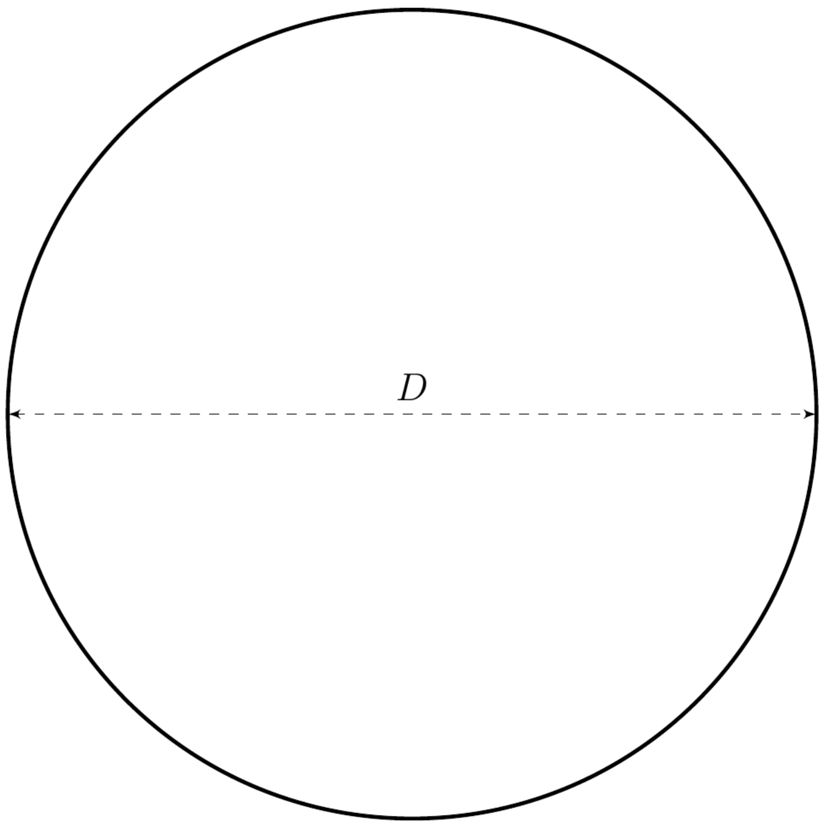
The image will look like a square with rounded corners. The straight part of the side has length a, the radius of curvature of the corners is $r = \dfrac{\alpha L}{2} = 4.4$ *cm*.



A4 0.50 pts

In the answer sheets, qualitatively give the image of the Sun at $a = 5$ mm. *Please indicate the dimensions that characterize this image.*

In this case, the image will differ slightly from a circle with diameter $D = \alpha L = 8.7\ cm$.



B1 **0.30 pts**

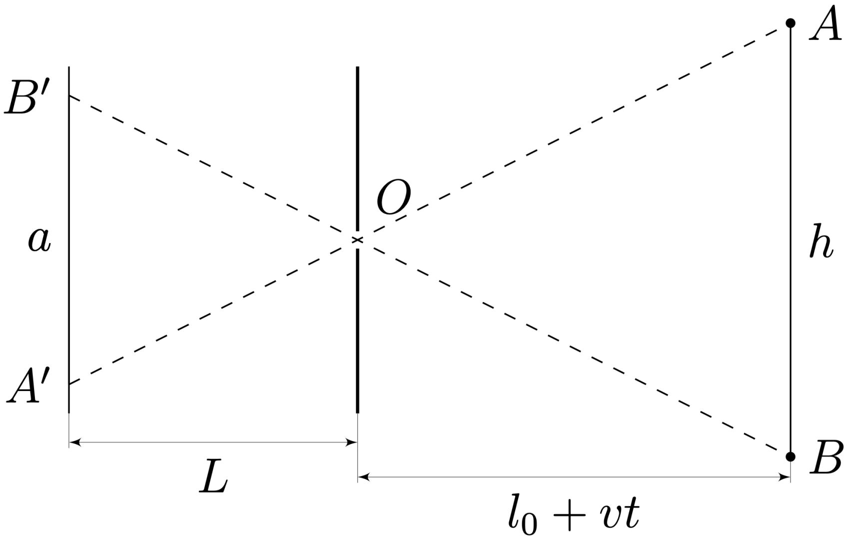
Find the dependence of the size a of a person's image on time t.

From the similarity of triangles AOB and $A'OB'$:

$$\frac{a}{h} = \frac{L}{l_0 + vt}.$$

Answer:

$$a = \frac{Lh}{l_0 + vt}.$$



B2 **0.70 pts**

The figure shows images of an adult human and the Sun (angular size of the Sun $\alpha = 0.5°$) at an unknown scale. Estimate how far away the adult was when the photograph was taken.

The angular sizes of the Sun and a person are of the same order, so we can assume that all the rays are paraxial and the sizes in the photograph are proportional to the angular sizes. From the figure, the ratio of the angular sizes of a person and the Sun:

$$\frac{h/l}{\alpha} = \frac{75mm}{30mm} = \frac{5}{2}.$$

For estimate take $h = 2m$:

$$l = \frac{2h}{5\alpha} \approx 90 \text{ m}.$$

Answer:

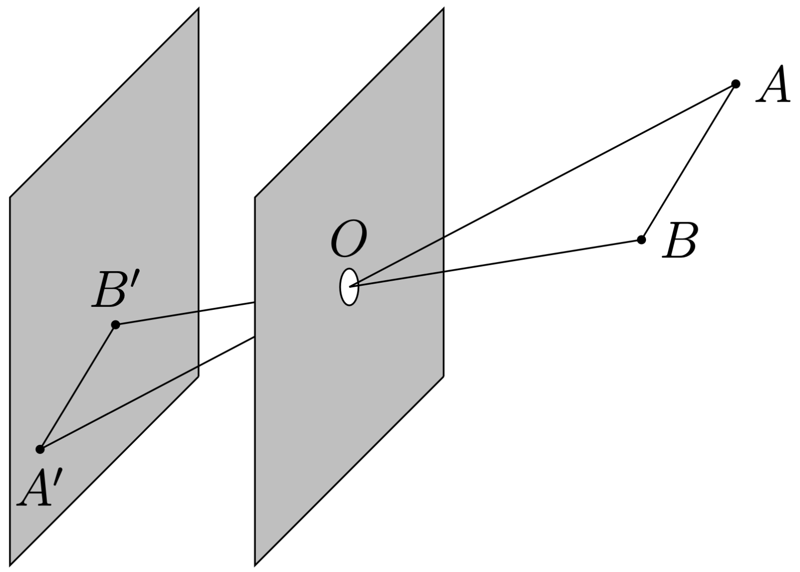
$$l \approx 90 \text{ m}$$

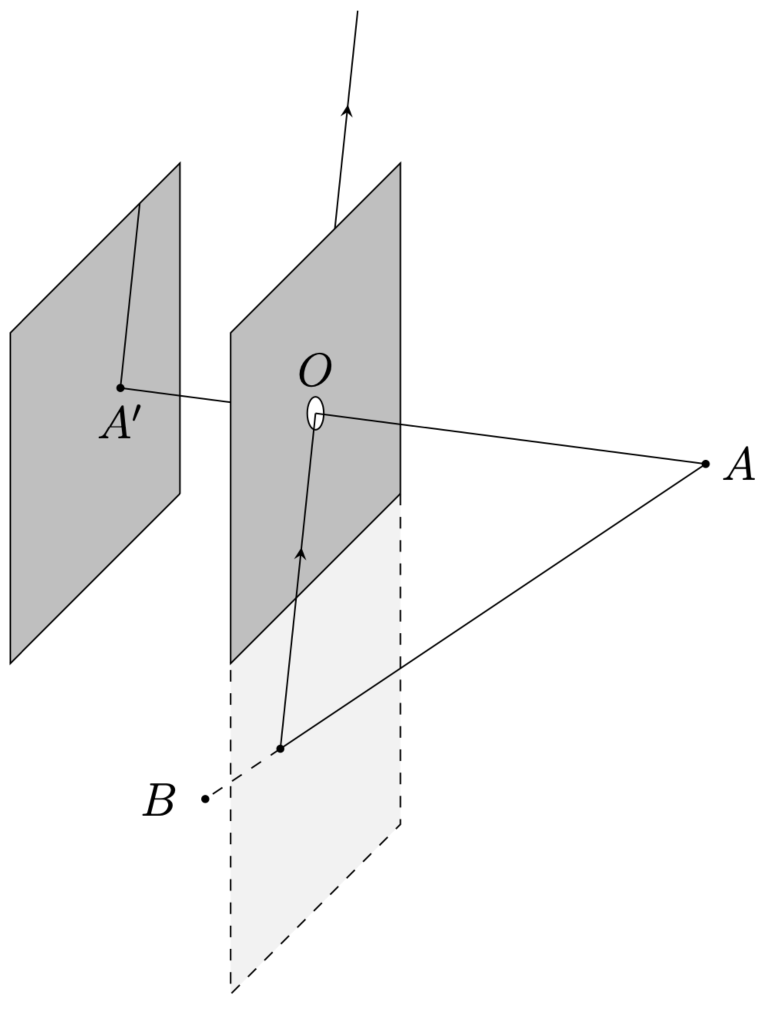
C1 **0.20 pts**

Prove that the image of a segment in a photograph will be a line segment or a ray.

Note that A, B, O and the image of the segment lie in the same plane, the image AB lies entirely on the straight line - the intersection of this plane with the plane of the screen. If A and B are in front of the front plane of the camera, then the image will be a line segment.

If A and B are on different sides of the front plane of the camera, then the image will be a ray.


A
B
O
B′
A′


O
A′
A
B

C2 **0.50 pts**

At what ratio between l_0 and a will all the flag points be depicted on the screen at any time, if we neglect the finiteness of the screen dimensions? In all subsequent paragraphs of this part of the problem, consider the resulting condition fulfilled.

Some part of the flag will not be in the photo if it lies on the opposite side of the front plane of the camera. Let's consider what is happening in the projection onto a plane perpendicular to the plane of the flag and the front plane of the camera.

In order for the flag to be visible, it is always necessary that for any position of the point E (E is the center of the flag), the following must be fulfilled: $EC < EF$. The length of EC is constant:

$$EC = \frac{a\sqrt{2}}{2}.$$

EF is the hypotenuse of the right triangle EOF, therefore $EF = 2l_m$, where l_m is the length of the median of this triangle from the vertex O. The height of EOF from vertex O is l_0, and it cannot be greater than l_m:

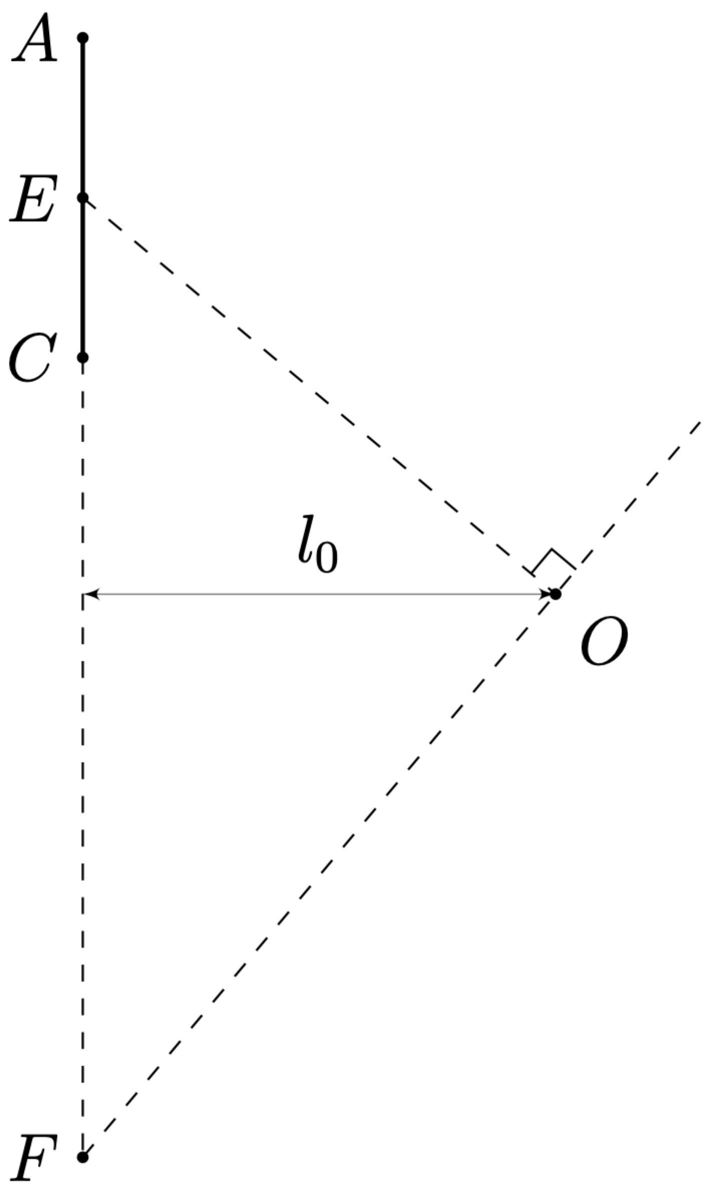
$$l_0 \leqslant l_m = \frac{EF}{2}.$$

That is $EF \geqslant 2l_0$ and equality holds when the triangle is considered isosceles. Thus, in order for the flag to always be visible, it can be seen, necessarily:

$$\frac{a}{\sqrt{2}} < 2l_0.$$

Answer:

$$a < 2\sqrt{2}l_0.$$



C3 **1.40 pts**

Find the coordinates (X, Y) of the vertices of the flag in the photograph depending on time.

The distance from the origin of coordinates XY to the image of a point is determined by the tangent of the angle between the camera axis and the direction to this point (for points A, B, C and D we denote these angles θ_A, θ_B, θ_C and θ_D, respectively). We can find the coordinates x, y and z of these points and get:

$$\tan\theta_B = \tan\theta_D = \frac{|x_B|}{z_B} = \frac{a/\sqrt{2}}{\sqrt{l_0^2 + v^2t^2}} = \frac{a}{\sqrt{2\,(l_0^2 + v^2t^2)}};$$

$$\tan\theta_A = \frac{|y_A|}{z_A} = \frac{\dfrac{a}{\sqrt{2}}\sin\xi}{\sqrt{l_0^2 + v^2t^2} + \dfrac{a}{\sqrt{2}}\cos\xi};$$

$$\tan\theta_C = \frac{|y_C|}{z_C} = \frac{\frac{a}{\sqrt{2}}\sin\xi}{\sqrt{l_0^2+v^2t^2} - \frac{a}{\sqrt{2}}\cos\xi}.$$

At the same time can express ξ through give quantity:

$$\cos\xi = \frac{vt}{\sqrt{l_0^2+v^2t^2}} \qquad \textit{and} \qquad \sin\xi = \frac{l_0}{\sqrt{l_0^2+v^2t^2}}.$$

We get the final answer:

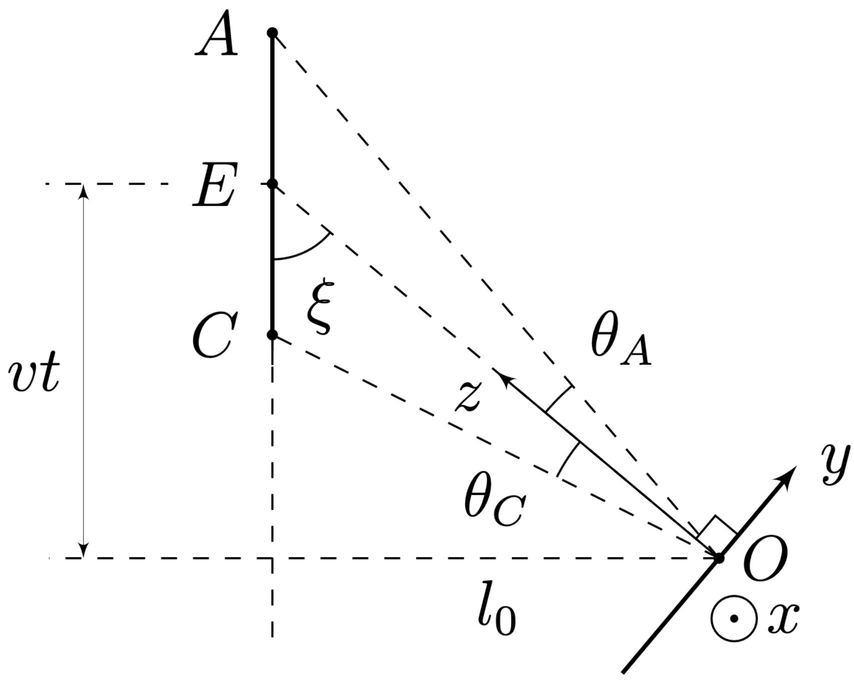
Answer:

$$X_A = 0 \qquad Y_A = \frac{\frac{a}{\sqrt{2}} l_0 L}{l_0^2+v^2t^2+\frac{a}{\sqrt{2}}vt}$$

$$X_B = -\frac{aL}{\sqrt{2\left(l_0^2+v^2t^2\right)}} \qquad Y_B = 0$$

$$X_C = 0 \qquad Y_C = -\frac{\frac{a}{\sqrt{2}} l_0 L}{l_0^2+v^2t^2-\frac{a}{\sqrt{2}}vt}$$

$$X_D = \frac{aL}{\sqrt{2\left(l_0^2+v^2t^2\right)}} \qquad Y_D = 0$$



C4 1.20 pts

The picture shows a photograph of the flag at an unknown scale. Determine at what point in time τ the photograph was taken.

From a photograph we can find the coordinates of the images of the vertices on an unknown scale, that is, we can find the tangents of the angles θ_A, θ_B, θ_C and θ_D, up to an unknown factor.

For convenience, we will use ξ in intermediate calculations.

$$\tan\theta_B = \frac{a/\sqrt{2}}{\sqrt{l_0^2+v^2t^2}} = \frac{\frac{a}{\sqrt{2}}\sin\xi}{l_0} = 8k$$

$$\tan\theta_A = \frac{\frac{a}{\sqrt{2}}\sin\xi}{\frac{l_0}{\sin\xi}+\frac{a}{\sqrt{2}}\cos\xi} = 3.6k$$

$$\tan\theta_C = \frac{\frac{a}{\sqrt{2}}\sin\xi}{\frac{l_0}{\sin\xi}-\frac{a}{\sqrt{2}}\cos\xi} = 7k$$

Divide $\tan\theta_B$ on $\tan\theta_A$ and $\tan\theta_C$.

$$\frac{8}{3.6} = \frac{1}{\sin\xi} + \frac{a}{\sqrt{2}l_0}\cos\xi$$

$$\frac{8}{7} = \frac{1}{\sin\xi} - \frac{a}{\sqrt{2}l_0}\cos\xi$$

Add, obtain value ξ.

$$\frac{8}{7}+\frac{8}{3.6} = \frac{2}{\sin\xi}$$

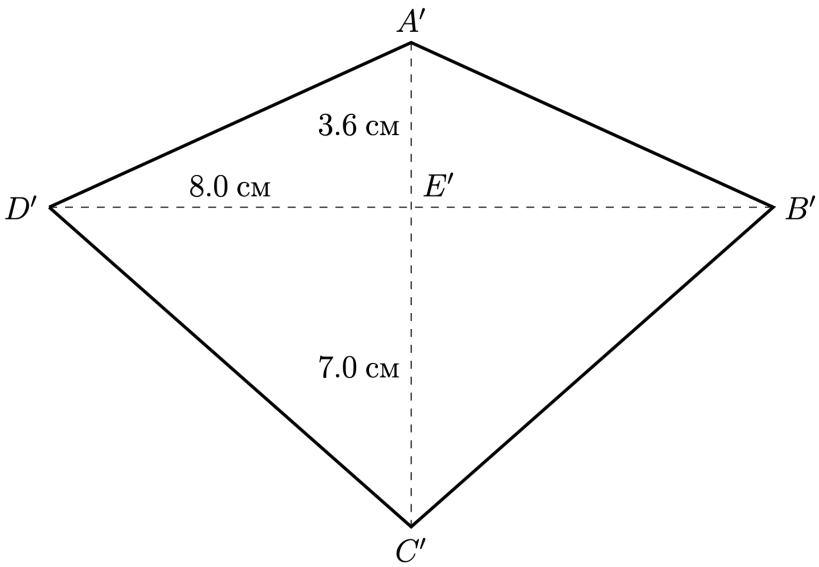
$$\sin\xi = \frac{l_0}{\sqrt{l_0^2+v^2t^2}} = 0.594$$

$$vt = 4.06 \text{ m}$$

We get answer:

Answer:

$$t = 4.06 \text{ s}$$



D1 — 0.10 pts

What is the angle θ? Express your answer in terms of R and h.

Answer:

$$\theta = \arccos \frac{R}{R+h}$$

D2 — 0.40 pts

Determine the coordinates x, y and z of the horizon point corresponding to angle φ. Express your answer using θ, R, φ.

The coordinate y of the horizon points does not depend on φ and it is equal to:

$$y = -\frac{R}{\cos\theta} + R\cos\theta.$$

Coordinate x and z equal:

$$x = R\sin\theta\sin\varphi;$$

$$z = R\sin\theta\cos\varphi.$$

Answer:

$$x = R\sin\theta\sin\varphi$$

$$y = -\frac{R}{\cos\theta} + R\cos\theta$$

$$z = R\sin\theta\cos\varphi$$

D3 — 0.20 pts

Prove that the coordinates X, Y are related to the coordinates x, y and z by the relations:

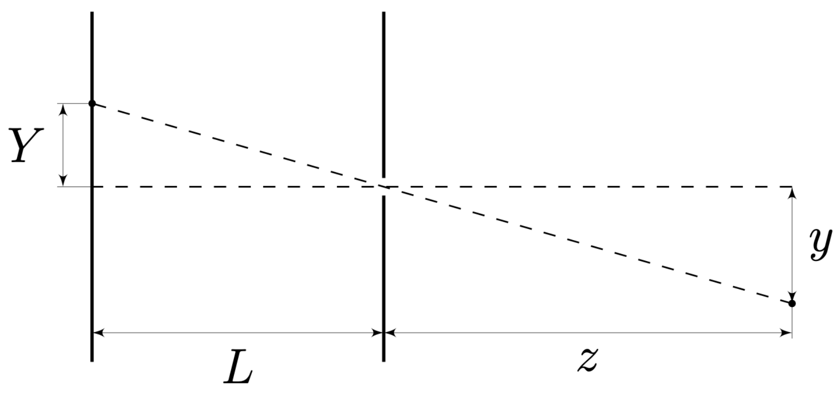
$$\frac{X}{x} = \frac{Y}{y} = \frac{L}{z}.$$

Let's consider what is happening in the projection onto yz.
From the similarity of triangles we get:

$$\frac{Y}{y} = \frac{L}{z}.$$

Similarly in projection on xz obtain:

$$\frac{X}{x} = \frac{L}{z}.$$



D4 0.70 pts

Show that the image of the horizon on the screen is described by the following equation:

$$\frac{Y^2}{A^2} - \frac{X^2}{B^2} = 1,$$

where $A, B > 0$. Express A and B through L and θ. Note: following expression may turn out useful:

$$\frac{1}{\cos^2\varphi} = 1 + \tan^2\varphi.$$

Note: The following expression may be useful:

$$\frac{1}{\cos^2\varphi} = 1 + \tan^2\varphi.$$

Using the results obtained earlier, we express X and Y in terms of L, θ and φ.

$$X = \frac{xL}{z} = L\tan\varphi$$

$$Y = \frac{yL}{z} = L\frac{-\dfrac{1}{\cos\theta} + \cos\theta}{\sin\theta\cos\varphi} = -\frac{L\tan\theta}{\cos\varphi}$$

Raise Y in square and express through X:

$$Y^2 = \frac{L^2\tan^2\theta}{\cos^2\varphi} = L^2\tan^2\theta + X^2\tan^2\theta.$$

$$\frac{Y^2}{(L\tan\theta)^2} - \frac{X^2}{L^2} = 1$$

Answer:

$$A = L\tan\theta$$

$$B = L$$

D5 1.00 pts

Determine from what height h the photograph was taken.

Let's use the formula from the previous paragraph of the problem. Let's substitute two points from the graph into it: $(0, 7k)$ and $(40k, 10k)$, where k is an unknown factor responsible for the scale.

$$\frac{(7k)^2}{A^2} = 1 = \frac{(10k)^2}{A^2} - \frac{(40k)^2}{B^2}$$

$$\tan^2\theta = \frac{A^2}{B^2} = \frac{10^2 - 7^2}{40^2}$$

Find θ and substitute in expression through h and R.

$$\theta = 10.1^\circ$$

$$\frac{h}{R} = \frac{1}{\cos\theta} - 1$$

Answer:

$$h = 100 \text{ km}$$

D6 — 0.50 pts

Determine the angle θ'. Express your answer in terms of R, n_0 and h.

The Earth's atmosphere is spherically symmetrical, therefore, when a beam moves in the atmosphere, the expression $n(r) \cdot r \cdot \cos\theta(r)$ is preserved, where r is the distance to the center of the earth, $n(r)$ is the refractive index of air at this point, $\theta(r)$ is the angle of direction of the ray with the horizon. Let's substitute the values at the surface of the earth and at the height h.

$$n_0 R = (R+h)\cos\theta'$$

Answer:

$$\theta' = \arccos\frac{n_0 R}{R+h}$$

D7 — 0.60 pts

Taking into account the smallness of $n_0 - 1$, determine by what amount Δh the real photographing height differs from the one you obtained in point D5. Express the answer in terms of $n_0 - 1$, R and h, and calculate its numerical value.

Let's write a new expression for θ and equate it to the old one, since the angle θ is determined from the graph and its value in the calculations will be the same.

$$\frac{n_0 R}{R+h+\Delta h} = \frac{R}{R+h}$$

$$(R+h)(n_0-1) = \Delta h$$

Answer:

$$\Delta h = (R+h)(n_0-1) = 1.77\ \text{km}$$

E1 — 0.30 pts

Determine h_{cr}, starting from which the horizon is completely visible when ascending.

The horizon is completely visible when the entire cone of rays from the horizon lies on one side of the front plane of the camera. This condition is satisfied at $2\psi < 90°$. Moreover, $\theta = 90° - \psi$, so the condition on θ:

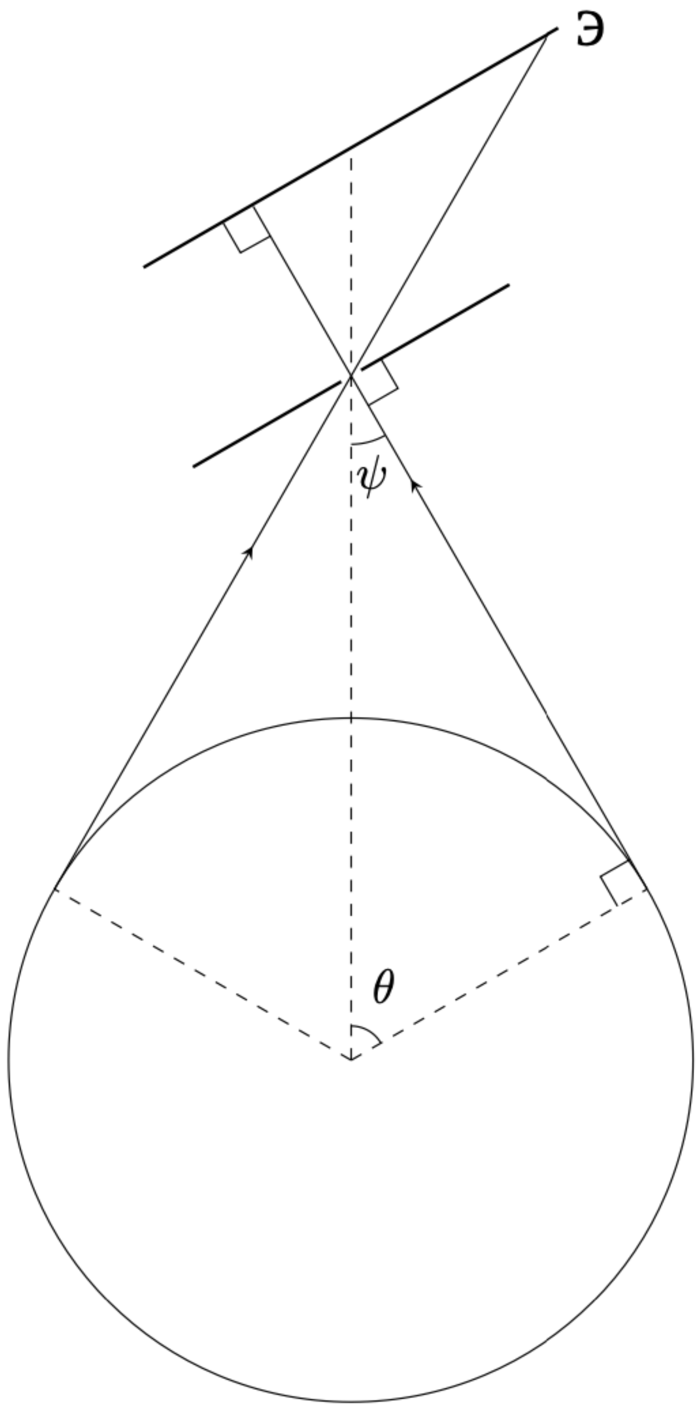
$$\theta > 45°.$$

This is equivalent to the following inequality:

$$R + h > \sqrt{2}R.$$

Answer:

$$h_{cr} = \left(\sqrt{2}-1\right)R$$



E2 **1.10 pts**

Determine the dependence of the visible part of the horizon η on the value h/R. Plot a graph of this relationship. Note: The angle θ shown in the figure may be useful for intermediate calculations.

Note: The angle θ shown in the figure may be useful for intermediate calculations.

Let's consider the section of the horizon circle by the front plane of the camera, when the horizon is not completely visible.

Note that the required value is:

$$\eta = \frac{2\pi - 2\gamma}{2\pi} = 1 - \frac{\gamma}{\pi}.$$

Height triangle on first photograph equal $\dfrac{R}{\cos\theta} - R\cos\theta$. Express Δ through R, θ.

$$\Delta = \left(\frac{R}{\cos\theta} - R\cos\theta\right)\tan\left(90^\circ - \psi\right) = \left(\frac{R}{\cos\theta} - R\cos\theta\right)\tan\theta = R\cos\theta\tan^3\theta$$

Express γ:

$$\gamma = \arccos\frac{\Delta}{r} = \arccos\frac{R\cos\theta\tan^3\theta}{R\sin\theta} = \arccos\left(\tan^2\theta\right).$$

The visible part of the horizon in this case:

$$\eta = 1 - \frac{\arccos\left(\tan^2\theta\right)}{\pi} = 1 - \frac{1}{\pi}\arccos\left(\left(1 + \frac{h}{R}\right)^2 - 1\right).$$

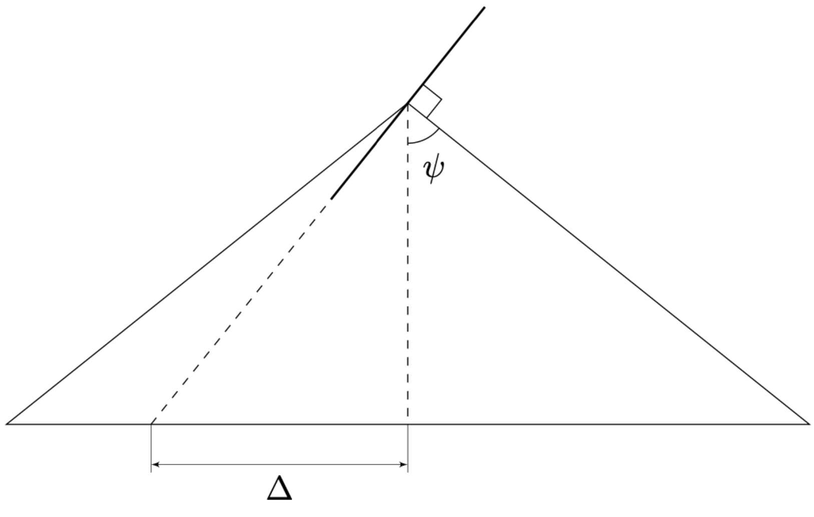
Note, that this formula is meaningful for $\theta < 45^\circ$, that is for $h < h_{cr}$. For $h \geqslant h_{cr}$ the horizon is fully visible, therefore $\eta = 1$.

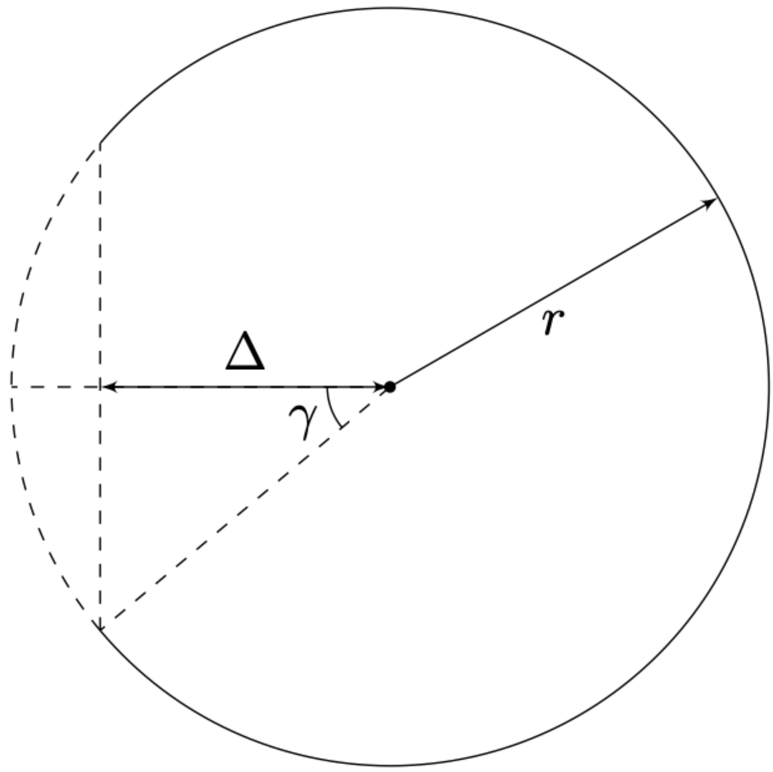
Answer: At $\dfrac{h}{R} < \sqrt{2} - 1:$

$$\eta = 1 - \frac{1}{\pi}\arccos\left(\left(1 + \frac{h}{R}\right)^2 - 1\right).$$

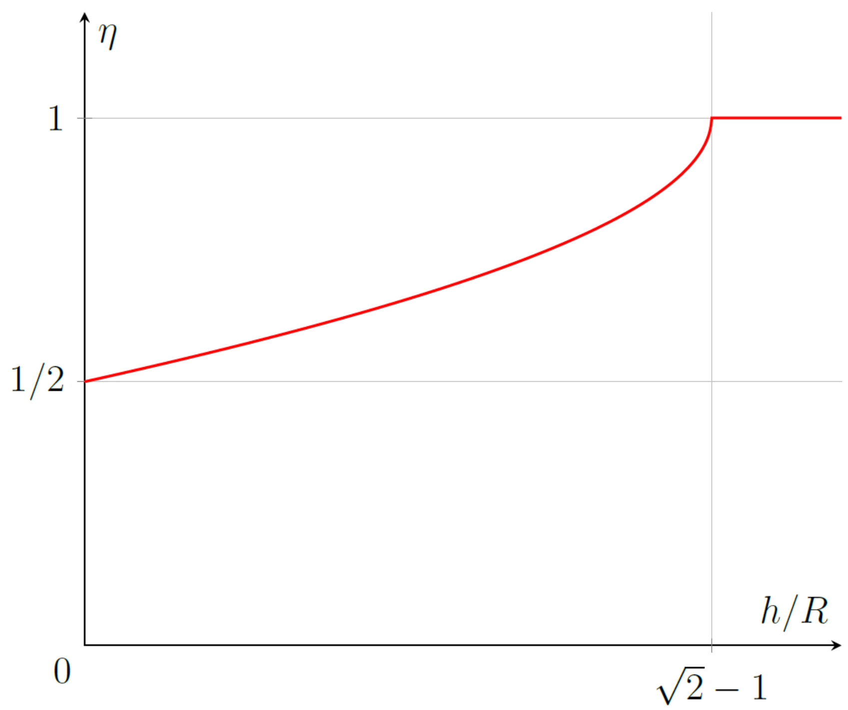
For $\dfrac{h}{R} \geqslant \sqrt{2} - 1:$

$$\eta = 1.$$





r
Δ
γ



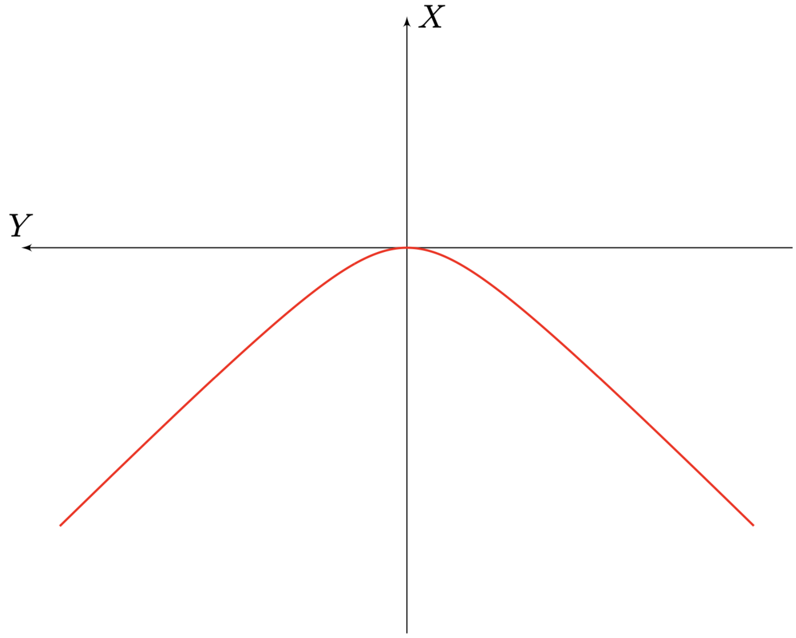
η
1
$1/2$
0
h/R
$\sqrt{2}-1$

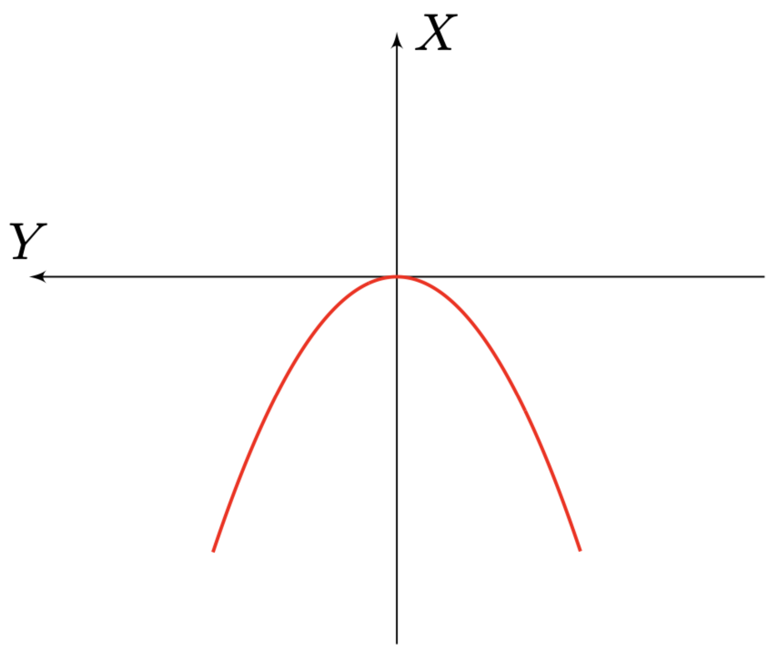
E3 **0.60 pts**

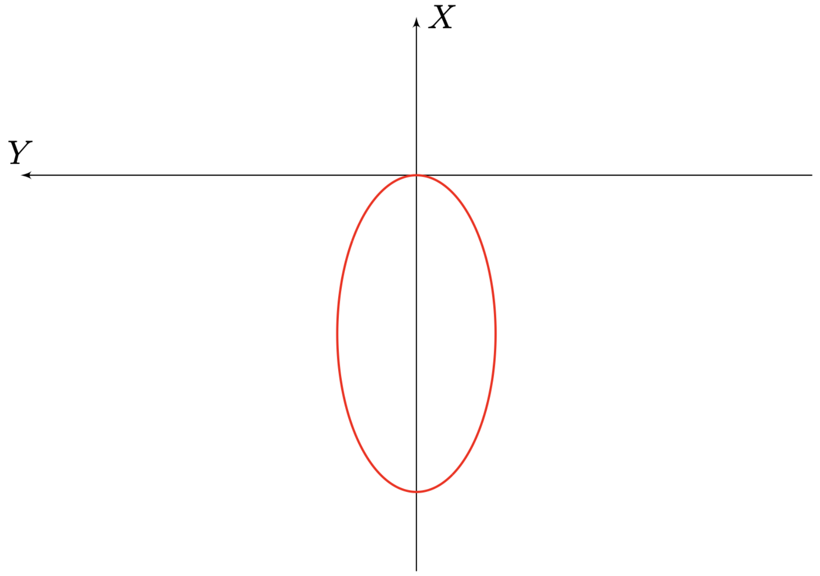
Draw high-quality images of the horizon at $h < h_{cr}$, $h = h_{cr}$ and $h > h_{cr}$. Determine the types of curves obtained when photographing from different heights.

Note that the image is a conic section. In the notation of the condition $\alpha = \psi = 90° - \theta$. In this case, the image will be an ellipse at $\theta > 90° - \theta$, a parabola at $\theta = 90° - \theta$ and a hyperbola at $\theta < 90° - \theta$. Substituting h, we get the answer.

Answer: At $h < h_{cr}$ the image is a hyperbole.






X
Y